

Answer Key: Assessment 1 Functions Practice

75 minutes | 100 points | Keep this copy separate from the student exam.

Section A: Function Basics

1. Read a function from a table (5 points)

Yes. The repeated x -value, 0, is paired with the same y -value, -1, so the relation is a function. The range is $\{-1, 4, 5\}$.

Scoring: 2 points for identifying the relation as a function; 2 for explaining that the repeated x -value has the same y -value; 1 for the range.

2. Catch the missing domain restriction (8 points)

$(f/g)(x) = \frac{\sqrt{5-x}}{x^2-1}$. The radical requires $x \leq 5$. The denominator is zero at $x = -1$ and $x = 1$, so both inputs must be excluded. Domain: $(-\infty, -1) \cup (-1, 1) \cup (1, 5]$.

Scoring: 2 points for the quotient; 2 for the radical restriction; 2 for identifying both denominator zeros; 2 for the interval domain.

3. Find and compare two compositions (10 points)

$B(A(x)) = (2x - 1)^2 + 3 = 4x^2 - 4x + 4$. Also, $B(A(4)) = B(7) = 52$, while $A(B(4)) = A(19) = 37$.

Scoring: 4 points for finding and simplifying $B(A(x))$; 3 for $B(A(4)) = 52$; 3 for $A(B(4)) = 37$.

Section B: Transformations and Piecewise Functions

4. Graph both orders of a composition (10 points)

The reference graph has vertices $(-4, -2)$, $(-2, 2)$, $(1, 0)$, $(4, 3)$, and $g(x) = 2x - 2$. The vertices of $f(g(x))$ are $(-1, -2)$, $(0, 2)$, $(\frac{3}{2}, 0)$, $(3, 3)$. The vertices of $g(f(x))$ are $(-4, -6)$, $(-2, 2)$, $(1, -2)$, $(4, 4)$. Connect each set of points in order using the requested line styles.

Scoring: 4 points for the vertices of $f(g(x))$; 4 for the vertices of $g(f(x))$; 2 for connecting the graphs correctly and marking endpoints.

5. Evaluate and graph a piecewise rule (8 points)

For $x = 0$, use the first rule: $p(0) = 2$. For $x = 1$ and $x = 3$, use the second rule: $p(1) = 3$ and $p(3) = 5$. The first branch ends with an open circle at $(1, 1)$. The second branch begins with a closed circle at $(1, 3)$ and follows the ray $y = x + 2$ to the right.

Scoring: 3 points for the three values; 2 for the open and closed boundary points; 3 for the graph and correct branch selection.

6. Graph a square-root translation (10 points)

The rule is $y = \sqrt{x + 4} - 3$. The starting point is $(-4, -3)$. The graph also passes through $(-3, -2)$, $(0, -1)$, $(5, 0)$ and continues to the right. The domain is $[-4, \infty)$ and the range is $[-3, \infty)$. Since $\sqrt{-3 + 4} - 3 = -2$, the point $(-3, -2)$ is on the graph.

Scoring: 2 points for the rule; 4 for the graph with its starting point and three additional points; 2 for the domain and range; 2 for checking the given point.

Section C: Polynomial Structure and Graphs

7. Recover a polynomial from its graph (7 points)

The graph crosses at $x = 2$ and touches at $x = -1$, so the factored form is $p(x) = a(x - 2)(x + 1)^2$. Its y-intercept is $(0, -1)$, so $-1 = a(-2)(1)$ and $a = 1/2$. Thus $p(x) = \frac{1}{2}(x - 2)(x + 1)^2$. The relative maximum is $(-1, 0)$ and the relative minimum is $(1, -2)$. It is increasing on $(-\infty, -1) \cup (1, \infty)$ and decreasing on $(-1, 1)$.

Scoring: 2 points for intercepts and multiplicity; 2 for using the y-intercept to determine the leading factor; 1 for the equation; 1 for the intervals; 1 for both extrema.

8. Choose the polynomial that fits the graph behavior (10 points)

Choice A fits: $(x + 2)(x - 1)^2$ has a simple zero at -2, so it crosses there, and a double zero at 1, so it touches and turns there. It is a positive cubic: the left end falls and the right end rises. Choice B has the multiplicities reversed and a negative leading coefficient; C has even multiplicity at both zeros and even degree; D has the right zeros and multiplicities but the wrong end behavior.

Scoring: 2 points for selecting A; 2 for matching zeros and multiplicities; 3 for degree, leading coefficient, and end behavior; 3 for explaining why the other choices do not fit.

9. Check a classmate's graph analysis (10 points)

The domain $[-5, 5]$, range $[-3, 4]$, and zeros $-4, 0, 4$ are correct. The graph is increasing on $(-3, 2)$ and decreasing on $(-5, -3) \cup (2, 5)$. The relative minimum is $(-3, -3)$ and the relative maximum is $(2, 4)$; extrema should be written as points, not only y-values.

Scoring: 2 points for checking domain and range; 2 for checking zeros; 3 for correct increasing/decreasing intervals; 3 for extrema as coordinate points.

Section D: Function Models and Transformations

10. Choose a valid composition using units (12 points)

The after-tax cost is $T(D(d)) = 1.08(9d + 12) = 9.72d + 12.96$. The rental charge must be calculated first, then the tax applied. For 4 days, $D(4) = 48$ dollars and $T(48) = 51.84$ dollars. Reversing the functions would feed a dollar amount into D , whose input is a number of rental days, so it would not model this situation.

Scoring: 4 points for selecting and explaining $T \circ D$; 3 for simplifying the composition; 3 for the \$51.84 total; 2 for explaining why the reversed order is not meaningful.

11. Keep the original domain when composing (10 points)

For $g(f(x))$, $(\sqrt{x+1})^2 - 1 = x$, but the formula for f requires $x+1 \geq 0$. Its domain is $[-1, \infty)$; simplifying does not restore values where the original square root was undefined. For $f(g(x))$, $\sqrt{(x^2-1)+1} = \sqrt{x^2} = |x|$. Since $x^2 \geq 0$ for every real x , its domain is $(-\infty, \infty)$.

Scoring: 4 points for $g(f(x)) = x$ and its restricted domain; 2 for explaining why simplification does not widen that domain; 2 for $f(g(x)) = |x|$; 2 for its all-real domain and comparison.